

Appendix: Exact Maximum Flexi-Clique Computation under Non-Hereditary Feasibility and Connectivity Constraints

A Hardness proof of maximum Flexi-clique problem

This section establishes the computational hardness of the Maximum Flexi-clique problem in two steps. Appendix A.1 proves NP-hardness by a reduction from the k -clique problem, and Appendix A.2 strengthens this result by showing that the problem admits no constant-factor approximation unless $P = NP$.

A.1 NP-hardness

Both reductions in this section rely on the same parameter choice, which makes the Flexi-clique condition collapse to the clique condition; we first record this property.

LEMMA 1. *Let $n \geq 3$ and $\tau = 1 - \frac{1}{n^2}$. Then for every integer $2 \leq x \leq n$, $\lfloor x^\tau \rfloor = x - 1$.*

PROOF. Since $\tau < 1$, we have $x^\tau < x$. For the other direction,

$$\ln\left(\frac{x}{x-1}\right) = -\ln\left(1 - \frac{1}{x}\right) \geq \frac{1}{x}. \quad (1)$$

Moreover, because $x \leq n$ and $\ln x < n$,

$$\frac{\ln x}{n^2} < \frac{1}{x}. \quad (2)$$

Consequently,

$$\ln\left(\frac{x}{x-1}\right) > \frac{\ln x}{n^2}, \quad (3)$$

which is equivalent to

$$\left(1 - \frac{1}{n^2}\right) \ln x > \ln(x-1). \quad (4)$$

Hence $x-1 < x^\tau < x$, and therefore $\lfloor x^\tau \rfloor = x-1$. $\square$

THEOREM 1. *The Flexi-clique decision problem is NP-hard.*

PROOF. We prove NP-hardness by reduction from the well-known k -clique problem, which is NP-complete in its decision form. Given an instance (G', k) of the clique decision problem, where $G' = (V, E)$ and $n = |V|$, we construct an equivalent instance (G, τ, k) of the Flexi-clique decision problem as follows. We may assume that $k \geq 2$ and $n \geq 3$, since instances with $k \leq 1$ or $n < 3$ can be decided directly. We set $G = G'$ and choose $\tau = 1 - \frac{1}{n^2}$; this parameter is rational and has a representation of $O(\log n)$ bits. By Lemma 1, for every node set H with $2 \leq |H| \leq n$, the Flexi-clique degree condition requires every node to be adjacent to all other nodes in H , and is therefore equivalent to the clique condition; moreover, every clique of size at least two is connected, so the connectivity requirement of Flexi-clique is automatically satisfied. The parameter k remains unchanged. If G' contains a clique H of size at least k , then H also forms a valid Flexi-clique in G , because each node has degree $|H| - 1$ within H . Conversely, if G contains a Flexi-clique H of size

at least k , the degree constraint forces H to be a clique, implying that G' contains a clique of size at least k . Since the transformation runs in polynomial time, the Flexi-clique decision problem is NP-hard. $\square$

COROLLARY 1. *The Maximum Flexi-clique problem is NP-hard.*

PROOF. We prove NP-hardness by reducing from the Flexi-clique decision problem (shown to be NP-hard in Theorem 1). Suppose we have a polynomial-time algorithm $\mathcal{A}$ that finds the maximum Flexi-clique in any instance (G, τ) . Given an instance (G, τ, k) of the Flexi-clique decision problem, we:

- (1) Use $\mathcal{A}$ to compute the largest Flexi-clique $H_{\max}$ and its size $|H_{\max}|$.
- (2) If $|H_{\max}| \geq k$, return yes; otherwise, return no.

Since it solves the NP-hard k -Flexi-clique problem in polynomial time, the Maximum Flexi-clique problem must also be NP-hard. $\square$

A.2 Constant factor inapproximability

THEOREM 2. *For every fixed constant $\rho > 1$, the maximum Flexi-clique problem is constant-factor inapproximable unless $P = NP$.*

PROOF. We give an approximation-preserving reduction from Maximum CLIQUE. Suppose, for contradiction, that there exists a polynomial-time ρ -approximation algorithm $\mathcal{A}$ for the Maximum Flexi-clique problem, where $\rho > 1$ is any fixed constant. Let $G = (V, E)$ be an instance of Maximum CLIQUE, and let $n = |V|$. We may assume that $n \geq 3$ and that G contains an edge, since the remaining cases can be decided directly. We retain the same graph and, as in the proof of Theorem 1, set $\tau = 1 - \frac{1}{n^2}$. By Lemma 1, the Flexi-clique degree condition on any node set H with $|H| \geq 2$ becomes $\delta(G[H]) \geq |H| - 1$, so the feasible Flexi-clique sets in (G, τ) are precisely the cliques of G with size at least two, and hence

$$\text{OPT}_{\text{flexi}}(G, \tau) = \text{OPT}_{\text{clique}}(G), \quad (5)$$

where $\text{OPT}_{\text{clique}}(G)$ denotes the size of a maximum clique in G . Applying $\mathcal{A}$ to (G, τ) returns a Flexi-clique H such that

$$|H| \geq \frac{\text{OPT}_{\text{flexi}}(G, \tau)}{\rho} = \frac{\text{OPT}_{\text{clique}}(G)}{\rho}. \quad (6)$$

Because every feasible Flexi-clique in this constructed instance is a clique, H is also a ρ -approximate solution to Maximum CLIQUE. This contradicts the known inapproximability of Maximum CLIQUE [7]. Therefore, no polynomial-time constant-factor approximation exists for the Maximum Flexi-clique problem unless $P = NP$. $\square$

B Correctness and complexity analysis

This section analyses EBA in three parts. Appendix B.1 analyses the running time: it establishes the per-node cost and the worst-case bound $O^*(2^n)$, explains why the pivot-based analyses that yield a base below 2 for the established cohesive subgraph models do not extend to Flexi-clique, and then proves a conditional bound below 2^n under bounded branching width. Appendix B.2 shows that the Flexi-clique model admits no size threshold that guarantees connectivity.

Appendix B.3 proves that **EBA** returns an exact maximum **Flexi-clique**. Throughout, $\tau \in (0, 1)$ is the degree parameter and $n = |V|$.

B.1 Running time analysis

Per-node cost and worst-case bound. By the completeness of CP-Branching and the soundness of the pruning rules established in the main paper, **EBA** computes a maximum **Flexi-clique** exactly. Its running time is the product of the number of search-tree nodes and the cost spent at each of them. Consider a search node $B = (S, C^r, C^{un}, D)$, and write $V_B = S \cup C^r \cup C^{un}$ for its remaining scope and E_B for the edge set of $G[V_B]$. The degree-based rule takes $O(|S| + |E_B|)$ time for the adjusted-degree computation and the follower cascade, the size-bound check takes $O(|S|)$ time, and the distance-based rule takes $O(|V_B| + |E_B|)$ time through incremental maintenance of distances. The work at each search node is therefore polynomial in $|V_B| + |E_B|$. Since every branch commits at least one node to either S or D , the search tree has $O(2^n)$ nodes, and the total running time is $O^*(2^n)$.

Why pivot-based analyses do not extend to Flexi-clique. For the established cohesive subgraph models—clique, k -plex, and γ -quasi-clique—branching algorithms with a worst-case base strictly below 2 have all been proven [3, 5, 6]. The common reason is that each of these models keeps every node’s required degree close to the subgraph size, which limits how many non-neighbours a node may have. A clique requires degree $|H| - 1$ and thus no non-neighbour; a k -plex requires degree $|H| - k$ and thus at most $k - 1$ non-neighbours per node; and a γ -quasi-clique requires degree $\gamma(|H| - 1)$ and thus at most a $(1 - \gamma)$ fraction of non-neighbours. In every case, the number of non-neighbours a node may have—which is the source of the branching width—is held to a constant or to a fixed fraction of the subgraph size, so selecting a pivot node bounds the branching width essentially for free, and a characteristic-root analysis yields a base below 2.

Flexi-clique weakens this leverage. Its degree requirement $\lfloor |H|^\tau \rfloor$ is sub-linear, falling ever further behind $|H|$ as the subgraph grows, so a node may be non-adjacent to almost the entire subgraph: the number of non-neighbours a node may have grows with $|H|$ rather than staying a constant or a fixed fraction below it. Since this quantity is precisely what a pivot exploits to bound the branching width, the pivot-based argument that succeeds for the established models no longer yields a base below 2 for **Flexi-clique**: pivoting still narrows the search, but it cannot cap the width by a constant. To the best of our knowledge, no analysis achieving a worst-case base below 2 is known for **Flexi-clique**, and whether such a bound is attainable remains open. This gap indicates why exact **Flexi-clique** computation is harder than for the established models, and why the graph scales and running times that **EBA** handles are not directly comparable to those of algorithms tailored to them. On real instances, however, the width-based pivot keeps the branching width small relative to n ; the remainder of this section quantifies the resulting gain, and Appendix C.1 measures the width empirically.

A conditional bound below 2^n . Recall that **EBA** explores a branch-and-bound tree whose search nodes have the form $B = (S, C^r, C^{un}, D)$. As the progress measure of a search node we use the number of undecided candidates, $\mu(B) = |C^r| + |C^{un}|$. In CP-Branching, the child B_i commits the candidates $v_1, \dots, v_i$ to S or D , and therefore reduces μ by at least i ; equivalently, if the explored children are ordered by rank, the j -th of them reduces μ by at least j . The following theorem shows that whenever the width-based pivot keeps the branching narrow, the base of the exponential drops strictly below 2.

THEOREM 3. *Suppose the width-based pivot bounds the number of explored children at every search node to at most w . Then **EBA** runs in $O^*(\alpha_w^n)$, where α_w is the largest root of*

$$x^{w+1} - 2x^w + 1 = 0, \tag{7}$$

and $\alpha_w < 2$ for every $w \geq 2$.

PROOF. Let $T(\mu)$ bound the size of the subtree rooted at a search node of measure μ . Since at most w children are explored and the j -th of them reduces μ by at least j , we obtain the recurrence $T(\mu) \leq \sum_{j=1}^w T(\mu - j)$. Substituting $T(\mu) = \alpha^\mu$ and dividing by $\alpha^{\mu-w}$ gives $\alpha^w = \sum_{l=0}^{w-1} \alpha^l = (\alpha^w - 1)/(\alpha - 1)$, and multiplying both sides by $(\alpha - 1)$ yields the characteristic equation (7). It remains to locate its largest root. Write $f(x) = x^{w+1} - 2x^w + 1$. Then $f(1) = 0$ and $f(2) = 2^{w+1} - 2 \cdot 2^w + 1 = 1 > 0$, while f attains a negative local minimum at $x = 2w/(w+1) < 2$ for every $w \geq 2$; hence the largest root α_w lies in the open interval $(1, 2)$. Since the per-node work is polynomial in the size of G , the total running time is $T(n) = O^*(\alpha_w^n)$. $\square$

For $w = 2, 3, 4$, the characteristic roots are $\alpha_w \approx 1.618, 1.839, 1.928$, respectively, and α_w approaches 2 as w grows. Keeping the width small is therefore what makes the bound meaningful, and this is precisely what the width-based pivot targets.

How the pivot bounds the width. Let

$$r_p = \bar{d}_{\text{margin}}(p) = \bar{d}_{\text{max}}(p) - \bar{d}_S(p) \quad (8)$$

denote the number of additional non-neighbours that the selected pivot p can tolerate before becoming saturated.

If $p \in S$, EBA orders the non-neighbours of p in C^r first, followed by its neighbours. Along the resulting sibling sequence, the number of non-neighbours of p included in the partial set increases monotonically. The first child contains no additional non-neighbour of p , and a child is saturated and therefore not explored once it contains $r_p + 1$ additional non-neighbours. Hence, at most

$$\min\{|C^r| + 1, r_p + 1\} \quad (9)$$

children are explored.

If $p \in C^r$, EBA places p first in the candidate ordering, followed by the non-neighbours of p in $C^r \setminus \{p\}$, and then by its neighbours. The first child excludes p , whereas every subsequent child contains p . Among the children containing p , the number of forced non-neighbours of p again increases monotonically, and exploration stops when this number reaches $r_p + 1$. Consequently, at most

$$\min\{|C^r| + 1, r_p + 2\} \quad (10)$$

children are explored, where the additional child accounts for the branch excluding p .

Thus, the effective branching width is controlled by the pivot margin up to an additive constant. When the selected margins remain small, the conditional base α_w in Theorem 3 is strictly below 2. The sub-linear degree requirement of Flexi-clique prevents a constant margin, and hence a constant w , from being guaranteed in the worst case; Appendix C.1 shows that the observed width nevertheless remains small in practice.

B.2 Absence of a size threshold for connectivity

Classical cohesive subgraph models either guarantee connectivity outright or recover it beyond a modest threshold: a clique is trivially connected; a k -plex H is connected—with diameter at most two—whenever $|H| \geq 2k - 1$ [2]; and a γ -quasi-clique is connected whenever $\gamma \geq 0.5$ [6]. The Flexi-clique model, in contrast, imposes only a size-dependent

degree constraint, and no analogous threshold exists: arbitrarily large node sets can satisfy the degree constraint while remaining disconnected. We make this precise below.

PROPERTY 1 (NO SIZE THRESHOLD FOR CONNECTIVITY). *For any $\tau \in (0, 1)$ and any integer h , there exists a node set $H \subseteq V$ with $|H| \geq h$ such that every node $v \in H$ satisfies $d(v, G[H]) \geq \lfloor |H|^\tau \rfloor$, yet $G[H]$ is disconnected.*

PROOF. Fix any $\tau \in (0, 1)$ and any integer h . Choose an integer s such that

$$2s \geq h \quad \text{and} \quad s - 1 \geq \lfloor (2s)^\tau \rfloor. \quad (11)$$

Such an s exists because $\frac{(2s)^\tau}{s} = 2^\tau s^{\tau-1} \rightarrow 0$ as $s \rightarrow \infty$. Hence, for all sufficiently large s , we have $(2s)^\tau \leq s - 1$, and therefore $\lfloor (2s)^\tau \rfloor \leq s - 1$. Now construct a graph $G = (V, E)$ consisting of two disjoint cliques C_A and C_B , each of size s , with no edges between them, and let $H = C_A \cup C_B$, so that $|H| = 2s \geq h$. Each node $v \in H$ has degree $s - 1$ in $G[H]$, and by (11) we have $s - 1 \geq \lfloor (2s)^\tau \rfloor = \lfloor |H|^\tau \rfloor$; hence every node satisfies the **Flexi-clique** degree constraint. However, since there is no edge between C_A and C_B , the induced subgraph $G[H]$ is disconnected. As h was arbitrary, the **Flexi-clique** model admits no size-based threshold that guarantees connectivity. $\square$

Consequently, connectivity cannot be inferred from the degree constraint and must be enforced explicitly during the search, which motivates the connectivity-preserving branching of **EBA**.

B.3 Exactness of EBA

This section gives the full proof of Theorem 3 of the main paper.

THEOREM 4 (EXACTNESS OF EBA). *Given a graph $G = (V, E)$ and a parameter $\tau \in (0, 1)$, **EBA** returns a maximum **Flexi-clique** of G .*

The proof rests on three facts: (i) without pruning, CP-Branching enumerates every connected induced subgraph of G exactly once (Proposition 1); (ii) every reduction of **EBA** discards only extension-infeasible nodes, i.e., nodes whose descendants cannot form a **Flexi-clique** larger than the incumbent F' ; and (iii) F' is a valid **Flexi-clique** at all times. Only the pivot-based sibling skipping of Section 5.3 is not already proved, so we establish its soundness first.

LEMMA 2 (SOUNDNESS OF SIBLING SKIPPING). *Let $B = (S, C^r, C^{un}, D)$ be a search node with $S \neq \emptyset$ and pivot p , and let $B_1, \dots, B_{k+1}$ be its children under the pivot ordering. If a child B_i with $p \in S_i$ satisfies $\bar{d}_{S_i}(p) > \bar{d}_{\max}(p)$, then every later child B_j ($j > i$) is extension-infeasible.*

PROOF. By sibling monotonicity, $S_i \subseteq S_j$ for every $j > i$, so $p \in S_j$ and every non-neighbour of p in S_i remains in S_j ; hence $\bar{d}_{S_j}(p) \geq \bar{d}_{S_i}(p) > \bar{d}_{\max}(p)$. Any descendant solution X of B_j contains S_j and lies in $H_{B_j} \subseteq H_B$. Thus X already forces more than $\bar{d}_{\max}(p)$ non-neighbours of p , while $|X| \leq |H_B|$ limits a **Flexi-clique** larger than F' to at most $\bar{d}_{\max}(p)$ of them. Hence no such X exists, and B_j is extension-infeasible. $\square$

PROOF OF THEOREM 4. For (iii), **EBA** initialises F' with the output of **FPA**, a valid **Flexi-clique** by construction (Section 4 of the main paper), and updates F' to S only when S is itself a **Flexi-clique** with $|S| > |F'|$; hence F' is always a valid **Flexi-clique**. For (ii), each reduction discards only extension-infeasible nodes: Rule 1 and its follower cascade by the definition of $\theta_{F'}$ (Equation (4) of the main paper), Rule 2 by Lemma 4 of the main paper, the size-bound termination by Lemma 2 of the main paper, and the sibling skipping by the lemma above.

Now let H^* be a maximum Flexi-clique of G , and suppose for contradiction that the returned F' satisfies $|F'| < |H^*|$. Since $G[H^*]$ is connected, Proposition 1 assigns H^* to a unique root-to-node path of the search tree. Every node on this path extends to H^* , which exceeds every incumbent; hence none is extension-infeasible, and by (ii) none is pruned. The search therefore reaches the node whose partial set is H^* and updates the incumbent to $|H^*|$, contradicting $|F'| < |H^*|$. Thus $|F'| = |H^*|$, and by (iii) F' is a valid Flexi-clique, so EBA returns a maximum Flexi-clique of G . $\square$

C Additional experiments

This section reports additional experiments: an empirical measurement of the branching width (Appendix C.1), a comparison with the exact maximum quasi-clique solver IterQC (Appendix C.2), and an ablation of the connectivity-preserving branching scheme (Appendix C.3).

Table 1. Per-node branching width on LFR graphs S_3 – S_6 as the graph size n grows ($\tau=0.5$).

Graph	S_3	S_4	S_5	S_6
n	10^3	10^4	10^5	10^6
W_{med}	3	3	2	2
W_{max}	6	17	11	8
$\alpha_{W_{\text{med}}}$	1.839	1.839	1.618	1.618

Table 2. Branching width on real networks ($\tau = 0.5$); W_{max}/n stays below 1%.

Dataset	AP	CD	CO	GW	DB	FD
n	16,046	39,577	90,303	196,591	317,080	1,070,376
W_{med}	44	231	20	255	34	3
W_{max}/n	0.00798	0.00644	0.00194	0.00130	0.00080	3.7×10^{-6}

C.1 Empirical branching width

Theorem 3 shows that if the width-based pivot keeps the number of explored children at every search node bounded by a constant w , then EBA runs in $O^*(\alpha_w^n)$ with $\alpha_w < 2$. The bound is conditional: as discussed in Appendix B.1, the sub-linear degree requirement of Flexi-clique prevents any constant w from being guaranteed in the worst case. We therefore measure the branching width empirically, instrumenting EBA to record, at every CP-Branching search node, the number of children it actually explores, and report the median width W_{med} and the maximum width W_{max} . The measurement uses two families of graphs: LFR synthetic networks [1] S_3 – S_6 , generated with mixing parameter $\mu = 0.3$, maximum degree 50, and maximum community size 10^3 , varying only the number of nodes so that S_i denotes the instance with $n = 10^i$ nodes for $i \in \{3, \dots, 6\}$; and the real networks in Table 2.

The key question is whether the width stays small relative to the graph size, and two findings show that it does. First, the width does not scale with n : as Table 1 reports, the median width on S_3 – S_6 stays at 2–3 while the number of nodes grows from 10^3 to 10^6 , corresponding to an effective base between $\alpha_2=1.618$ and $\alpha_3=1.839$. Second, on real networks (Table 2), the width remains a tiny fraction of n : the maximum width is at most a few hundred even on graphs with hundreds of thousands to millions of nodes. Empirically, the observed width appears to be governed more by the local degree structure and the resulting pivot margins than by the total graph size. Thus, while a constant worst-case width cannot be guaranteed, in practice the branching width stays orders of magnitude below n and does not grow with it, so EBA effectively explores a search tree far narrower than the trivial 2^n bound.

Table 3. Running time (s) of EBA and IterQC ($\tau, \gamma=0.5$).

Alg.	ED	PG	AP	CA	CD	BK	CR	CO	FC	GW	DB	AM	FD	YT
EBA	0.029	0.056	0.747	0.109	4.050	1.261	2.919	1.903	45.92	44.54	21.93	2.055	0.820	54.84
IterQC	0.034	2.100	0.519	1.021	0.553	0.653	T/O	0.258	4.028	1,007	1.153	0.863	1.175	7,034

Table 4. Sizes of the maximum Flexi-clique and the maximum quasi-clique ($\tau, \gamma=0.5$).

Alg.	ED	PG	AP	CA	CD	BK	CR	CO	FC	GW	DB	AM	FD	YT
EBA	80	194	993	168	360	561	1,370	255	9,024	1,443	899	38	15	2,208
IterQC	21	49	107	45	49	101	T/O	45	1,051	87	115	13	7	75

C.2 Comparison with exact maximum γ -quasi-clique search

We compare EBA with IterQC [4], a state-of-the-art exact solver for the maximum γ -quasi-clique problem. This comparison is not intended as a like-for-like algorithmic ranking, because the two methods solve different optimisation problems under different feasibility conditions. Instead, IterQC serves as a structural reference for exact cohesive-subgraph search under a linear-scaling degree threshold.

The two models differ in properties that directly affect exact search. The degree threshold of a γ -quasi-clique grows linearly with the subgraph size, whereas the Flexi-clique threshold $\lfloor |H|^\tau \rfloor$ grows sub-linearly. Moreover, for $\gamma \geq 0.5$, a γ -quasi-clique is connected and has diameter at most two [6]. These properties permit neighbourhood-based search restrictions without a separate connectivity-preserving mechanism. In contrast, the Flexi-clique degree condition does not imply connectivity for any fixed $\tau \in (0, 1)$, as shown in Appendix B.2; EBA therefore preserves connectivity explicitly through CP-Branching. We report IterQC only for $\gamma \geq 0.5$, which is the parameter range considered by IterQC.

Tables 3 and 4 report the running times and solution sizes of EBA and IterQC at the same numerical parameter values, 0.5, under a one-day time budget. EBA is faster on some instances, whereas IterQC is faster on others. The two models also return markedly different solution sizes, with Flexi-clique generally producing larger subgraphs under the tested settings. Consequently, the running-time results should be interpreted together with the distinct feasibility conditions and solution spaces of the two models, rather than as a direct performance ranking.

Table 5. Running time (s) of CP-Branching and SE under identical pruning ($\tau=0.5$).

Scheme	ED	FD	PG	AP	CD	CO	GW	DB
CP-Branching	0.03	0.82	0.06	0.75	4.05	1.90	44.5	21.9
SE	0.03	1.20	T/O	T/O	T/O	T/O	T/O	T/O

C.3 Contribution of connectivity-preserving branching

To isolate the contribution of CP-Branching, we compare it against SE, a set-enumeration branching scheme in the style of quasi-clique enumerators that includes each candidate in turn and recurses on the strictly later ones. Both schemes explore the same solution space under identical pruning rules, with τ fixed to 0.5; the only difference is the branching. Since SE does not distinguish candidates by reachability, it may extend a partial set with any remaining candidate, and therefore has to verify connectivity, on top of the degree constraints, for every partial set it forms; CP-Branching instead maintains connectivity by construction and never performs an explicit check.

Table 5 reports the running time. When the search space remaining after the initial pruning is narrow, the two schemes behave almost identically. Once substantial branching is required, however, the undifferentiated candidate set of SE makes its branches grow excessively long while the connectivity checks accumulate, and SE fails to terminate within the 10-minute budget on all six remaining datasets, whereas CP-Branching finishes each of them within a minute. This asymmetry explains why set-enumeration branching serves the previous models well—their hereditary structure or bounded diameter renders connectivity essentially free—but breaks down for Flexi-clique, which satisfies neither property (Appendix B.2); explicitly separating candidates by reachability, as CP-Branching does, is what keeps exact search tractable.

References

- [1] Andrea Lancichinetti, Santo Fortunato, and Filippo Radicchi. 2008. Benchmark Graphs for Testing Community Detection Algorithms. *Physical Review E* 78, 4 (2008), 046110.
- [2] Stephen B. Seidman and Brian L. Foster. 1978. A Graph-Theoretic Generalization of the Clique Concept. *Journal of Mathematical Sociology* 6, 1 (1978), 139–154.
- [3] Etsuji Tomita, Akira Tanaka, and Haruhisa Takahashi. 2006. The Worst-Case Time Complexity for Generating All Maximal Cliques and Computational Experiments. *Theoretical Computer Science* 363, 1 (2006), 28–42.
- [4] Hongbo Xia, Kaiqiang Yu, Shengxin Liu, Cheng Long, and Xun Zhou. 2025. Maximum Degree-Based Quasi-Clique Search via an Iterative Framework. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2*. 3285–3296.
- [5] Mingyu Xiao, Weibo Lin, Yuanshun Dai, and Yifeng Zeng. 2017. A Fast Algorithm to Compute Maximum k -Plexes in Social Network Analysis. In *Proceedings of the 31st AAAI Conference on Artificial Intelligence*.
- [6] Kaiqiang Yu and Cheng Long. 2023. Fast Maximal Quasi-Clique Enumeration: A Pruning and Branching Co-Design Approach. *Proceedings of the ACM on Management of Data* 1, 3 (2023), 1–26.
- [7] David Zuckerman. 2006. Linear Degree Extractors and the Inapproximability of Max Clique and Chromatic Number. In *Proceedings of the 38th Annual ACM Symposium on Theory of Computing*. 681–690.